



86 It

Say it once. Every screen knows.

Built on the AssemblyAI Voice Agent API. A cook says “86 salmon” into a headset. The moment the transcript lands, the kitchen display, the servers’ screen and the online ordering page all agree.

When a kitchen runs out, the news travels by shouting, **and the apps keep selling it.**

≈75%

of US restaurant traffic is off-premises. The screens are the storefront.

NRA, Off-Premises Trends 2025

Greasy gloves can't tap screens

1.5-3%

of US online orders were cancelled in 2023, and cancellations track the rating.

Otter, 2023 order data

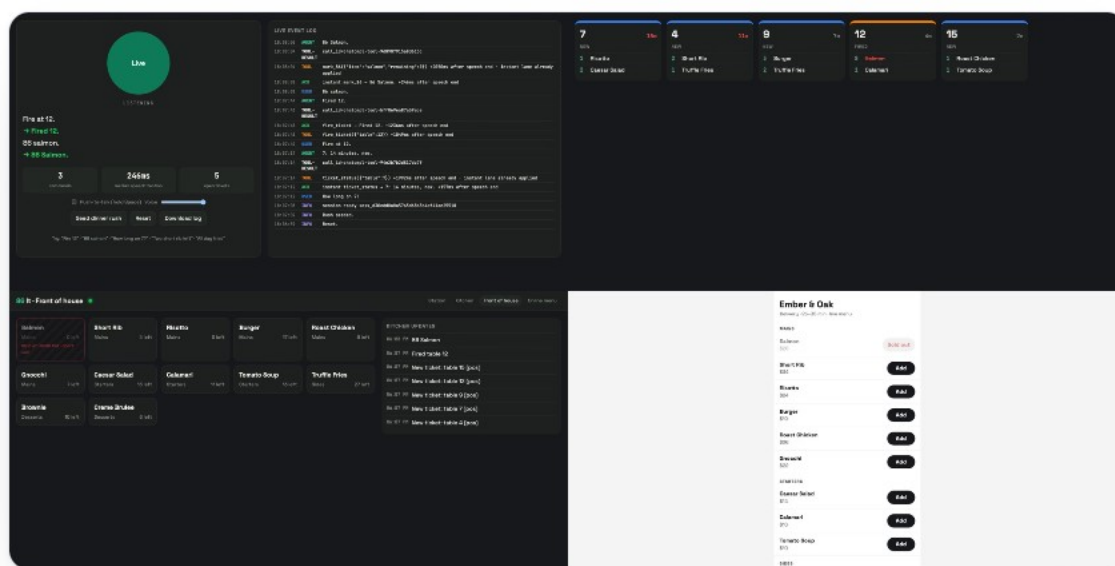
4.5★ → 4.0★

average rating below 2% cancellations vs above 8%. Two in a row can pause the store.

Otter, December 2023

Too loud for a voice assistant

Say “86 salmon” once and every screen changes in the same moment.



Kitchen display

Salmon greys out. The line stops cooking it.

Servers' screen

Front of house can no longer sell it.

Online menu

Delivery pages reject it the same instant.

One utterance → three screens
0 ms instant · 2.1 s LLM

Cooks speak kitchen slang; the system answers in one to five words.

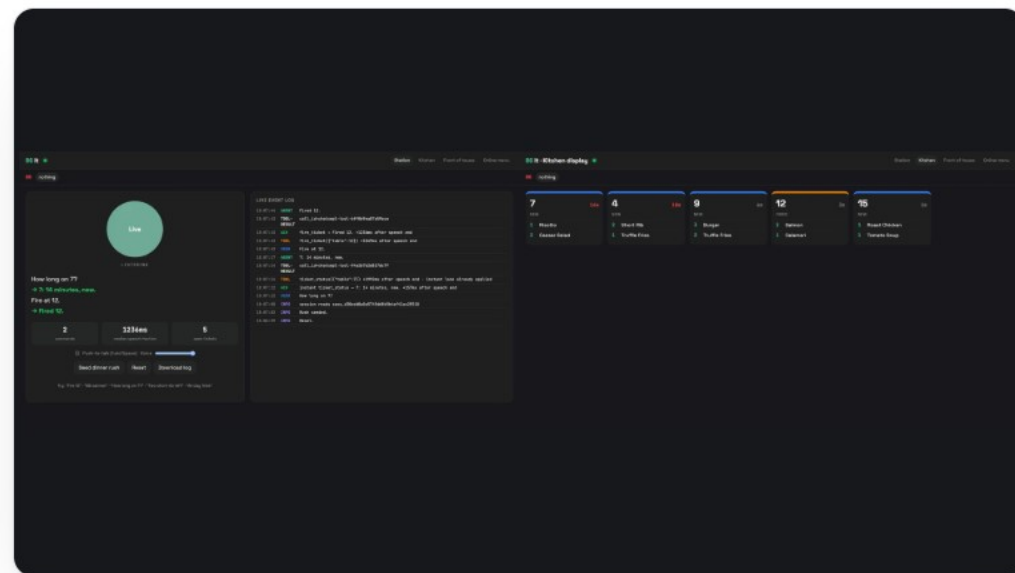
86 salmon	Greys out on KDS, servers' screen, menu.
two short rib left	Count updates everywhere, item flagged low.
fire 12 · bump 4 · hold 9	The ticket moves on the kitchen display.
how long on 7?	"Seven: 14 minutes." · "All day: three."
12 has a nut allergy	A red banner appears on ticket 12.
void 7 → "yes"	Asks "Void 7?" first. Nothing until "yes".
kitchen chatter	Silence: no tool call, no spoken reply.
ticket passes 10 min	Speaks unprompted: "Table 4, 12 minutes."

Enums restrict the LLM to menu items

Keyterms carry the slang

The same voice loop is **deployed and live**. This is what runs on stage.

- 0:20** “Fire twelve.” → ticket 12 turns orange.
- 0:40** “Eighty-six salmon.” → three screens grey out.
- 1:25** A teammate talks football. Nothing happens.
- 1:55** Late-ticket alert: “Table 4, 12 minutes.”



Turns orange mid-reply

One WebSocket carries the voice; **one Durable Object owns the state** and fans it out.

- 01 Station** browser + headset: 24 kHz PCM16 streams.
↓ `input.audio`, streamed as the cook speaks
- 02 AssemblyAI Voice Agent** STT, turn detection, LLM + 10 tools, TTS.
↓ `transcript` → instant lane acts; LLM call deduped
- 03 Worker + Durable Object** tokens, authoritative state, fan-out.
↓ `state broadcast to every open socket`
- 04 Every screen** kitchen display, servers' board, online menu.

Screens change on the broadcast, **before** the agent speaks.

Grammar first · LLM confirms

Temp token, key stays server-side

Every hard part of voice is an AssemblyAI feature we wired up, **not code we wrote.**

Voice Agent API

One WebSocket: STT, LLM, speech, turn detection.

Keyterms + menu enums

Slang transcribes; enums stop invented dishes.

min_latency + barge-in

Fast transcription; the cook can cut the agent off.

Temp tokens from /api/token

Inline session config

Prompt, 10 tools and keyterms in session.update.

Voice focus + near-field

Isolates the cook from a loud line, every time.

reply.create + resume

Unprompted alerts; survives a Wi-Fi blip.

Client tools declared inline

43 of 43 clips pass, 42 of 43 in noise.

43-clip eval	Clean	Pink noise -12 dB
Pass · LLM only	42 / 43	42 / 43
Pass · + instant lane	43 / 43	42 / 43
Chatter ignored	12 / 12	12 / 12
Speech → action (p50)	0 ms · LLM 2.1 s	0 ms · LLM 2.0 s
Spoken confirm (p50)	4.9 s	4.2 s

The one noisy miss was a mishearing, and it failed [safe](#).

22 unit tests and a protocol-faithful fake caught violations before we spent a credit.

22

unit tests over the kitchen reducer, fuzzy matching, void safety and alerts.

```
npm test · node --  
test
```

0

protocol violations across a full voice loop: tools, barge-in, chatter, resume.

```
GET :9901/stats →  
violations: []
```

86

sessions recorded with AssemblyAI's own recording and timeline.

```
recordings/ ·  
eval/results/
```

The mock flags protocol violations

On every recorded session

Per-location subscriptions, sold through **the POS vendors restaurants already run.**

Buyer

The owner or kitchen manager, for independents and groups.

Distribution

An add-on for POS/KDS vendors: Toast, Square, Lightspeed.

Next: multi-station kitchens · multi-location dashboards · manager analytics

Price

Per location per month, optional headset + tablet bundle.

Value

Fewer refunds, faster tickets, allergy flags, a timestamped 86 log.

What is honestly **not finished yet.**

Limits today

One restaurant per deployment; no auth.

The instant lane covers fixed phrasings only.

The POS is a mock.

Eval used synthesized speech.

Off-grammar speech takes 2-4 s; “Un-86 X” is flaky.

Next

Key the Durable Object by restaurant id.

Add auth to the demo screens.

An owner phone line: “what did we 86 tonight?”

Real voices and noise in the eval.

Toast / Square inventory via act().

Say it once. Every screen knows.

Live `86-it.rd5510.workers.dev` **Write-up** `docs/SUBMISSION.md`

Video `86-it-demo-final.mp4` · 2:16

Sending tool results immediately **cut two seconds of dead air** from every confirmation.

What the docs suggested

Wait for `reply.done` before sending `tool.result`: the agent may be speaking a transition.

What the timelines showed

That transition is silent. We send `tool.result` immediately: confirmations ~2 s sooner, zero errors in any recorded session.

Measured and reverted: `execution_mode hold · shorter prompt · min_silence 400 ms`

transcription_mode: min_latency **bought 0.6-1 s**
at the same accuracy.

balanced mode · 3-command A/B

2.85-4.3 s



min_latency · same 3 commands

2.2-3.7 s



Prompt fixes that landed: chatter gets silence, menu nicknames (“the fish” → salmon), number homophones, and a silent-reply gate.